

Numerical Methods for Elliptic Problems

Lecture Notes for the Winter Semester 2026/27

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September 10, 2026

Preface

These lecture notes are based on the original script by Herbert Egger. They have been adapted and revised by Michael Neunteufel for the course *Numerical Methods for Elliptic Problems* at Johannes Kepler University Linz in the winter semester 2026/27.

Corrections, modifications, and additional material in this version are the responsibility of the adapting author.

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Introduction

This course is about elliptic variational problems, stemming from boundary value problems for certain partial differential equations, and their systematic numerical approximation by the finite element method. Most of the theory will be developed for a simple system, i.e., the Dirichlet problem for the Poisson equation

$$\begin{aligned} -\operatorname{div}(a\nabla u) &= f && \text{in } \Omega, \\ u &= 0 && \text{on } \partial\Omega. \end{aligned}$$

The results and methods, however, carry over quite naturally to much more general problems, e.g., in heat and mass transfer, mechanics, fluid flow and electromagnetics, making the finite element method a very powerful tool for the simulation of complex physical processes in science and engineering.

During the lecture, we will touch on various aspects that are important for the development and analysis of discretization methods:

- Modelling: study the physical meaning of the equations and the mathematical tools underlying their derivation.
- Analysis: establish existence, uniqueness and stability of solutions.
- Discretization: develop tools for the systematic numerical approximation.
- Numerical analysis: analyse discretized problems and prove convergence.
- Implementation: study the basic approach to an efficient realization and derive complexity estimates for the numerical solution.

Literature. For preparation of this lecture, we used material from standard textbooks on finite element methods, in particular

- D. Braess: Finite Elements: Theory, Fast Solvers, and Applications in Solid Mechanics. Cambridge University Press, 2007.

as well as

- P. Ciarlet: The Finite Element Method for Elliptic Problems. SIAM, 2002.
- S. Brenner and L. R. Scott: The Mathematical Theory of Finite Element Methods. Springer 2008.
- A. Ern and J.-L. Guermond: Theory and Practice of Finite Elements. Springer, 2004.
- S. Bartels: Numerical Approximation of Partial Differential Equations. Springer 2016.

Further references for particular results will be given during the course.

1 Model problems and principles

Before we delve into the details of the mathematical analysis and numerical approximation of elliptic boundary value problems for partial differential equations, we want to highlight some physical principles and mathematical arguments used for their derivation and introduce some typical model problems that lie in the wider scope of the lecture.

1.1 Mass and heat transfer

We consider the transport of a chemical substance, a tracer, in the flow of a fluid, e.g., water or air. By the principle of *conservation of matter*, we obtain the balance equation (change of total amount = flow over boundary + sources/sinks)

$$\frac{d}{dt} \int_{\omega} u(x, t) dx = - \int_{\partial\omega} \vec{q}(x, t) \cdot \vec{n}(x) ds(x) + \int_{\omega} s(x, t) dx,$$

which holds for every control volume ω under consideration. Here u is the concentration of the substance, $\vec{q}$ the flux, $\vec{n}$ the outer normal on $\partial\omega$, and s a source term. Under steady state conditions, all quantities can be assumed independent of time. Using the Gauß-integral law

$$\int_{\omega} \operatorname{div}(\vec{q}) dx = \int_{\partial\omega} \vec{q} \cdot \vec{n} ds$$

and dropping the left hand side in the first equation, we arrive at

$$\int_{\omega} \operatorname{div}(\vec{q}) dx = \int_{\omega} s dx,$$

which still holds for all control volumes ω under consideration. This can be turned into a partial differential equation by the following result.

Lemma 1.1 (Fundamental lemma 1).

Let $\Omega \subset \mathbb{R}^d$ be some open set and $f \in C(\overline{\Omega})$. Then

$$\int_{\omega} f(x) dx = 0 \quad \forall \omega \subset \Omega \quad \Leftrightarrow \quad f(x) = 0 \quad \forall x \in \Omega.$$

It suffices to restrict the control volume ω to closed balls lying in Ω .

The proof is left as an exercise. Note that a similar statement holds for integrable $f \in L^1(\Omega)$. Then the right identity holds for almost every (a.e.) $x \in \Omega$.

By use of the fundamental lemma, we directly obtain the following balance equation in differential form

$$\operatorname{div}(\vec{q}) = s \quad \text{in } \Omega,$$

where $\Omega \subset \mathbb{R}^d$ is the *computational domain* which we consider. In order to recover information about the concentration u , we have to prescribe how the flux $\vec{q}$ and the source density s depend on u . For this we use the *constitutive equations*

$$\vec{q} = -a\nabla u + \vec{b}u \quad \text{and} \quad s = f - cu,$$

where a is a diffusion coefficient, $\vec{b}$ the flow velocity of the fluid, f an external source, and c a reaction coefficient. The first term in the flux amounts to diffusion (=Fick's law) and the second to transport (=Transport theorem). The source term says that substance is either injected/subtracted directly or degrades through some chemical process.

By inserting the constitutive laws into the balance equation, we arrive at the *diffusion-convection-reaction* equation

$$-\operatorname{div}(a\nabla u - \vec{b}u) + cu = f \quad \text{in } \Omega. \quad (1.1)$$

In order to fully describe the concentration profile in steady state, we additionally require *boundary conditions*, e.g.

$$u = g \quad \text{on } \Gamma_D \quad (1.2)$$

$$\vec{n} \cdot (a\nabla u - \vec{b}u) = h \quad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \quad (1.3)$$

The first (=Dirichlet) condition directly prescribes the concentration, while the second (=Neumann) condition fixes the normal flux at the boundary. Note that we prescribe exactly one condition at every point of the boundary.

Model problem

The system (1.1)–(1.3) makes up a *boundary value problem* for a *linear elliptic partial differential equation of second order* (short: an elliptic bvp) which describes the distribution of a chemical substance in the steady flow of a fluid. By the very same arguments, we could also consider the transfer of heat and arrive at the stationary heat equation. If we assume $\Gamma_N = \emptyset$, $\vec{b} = 0$, $c = 0$, and $g = 0$, the above system further simplifies to

$$-\operatorname{div}(a\nabla u) = f \quad \text{in } \Omega, \quad (1.4)$$

$$u = 0 \quad \text{on } \partial\Omega, \quad (1.5)$$

which will serve as the basic model problem for most of our considerations. Another interesting case occurs, if we neglect diffusion and set $a = 0$. Then we obtain a pure transport problem, which is a linear hyperbolic differential equation of first order. We will come back to this problem at a later stage.

1.2 Minimal surface problem

We consider a thin membrane which is stretched over a fixed frame, e.g., a wire loop. The position of the frame is given by a height function $g : \partial\Omega \rightarrow \mathbb{R}$, where $\Omega \subset \mathbb{R}^2$ is the projection of the area enclosed by the frame onto a flat surface. Now let $u : \Omega \rightarrow \mathbb{R}$ denote the vertical position of the membrane. We assume that in equilibrium, the membrane takes the form of minimal surface area, i.e.,

$$A(u) := \int_{\Omega} \sqrt{1 + |\nabla u(x)|^2} dx \rightarrow \min_{u: u|_{\partial\Omega} = g}.$$

Some restrictions on the possible deformations (=deflections) described by u are, of course, required to ensure that the integral is well-defined and, eventually, to guarantee the existence of a minimizer. If u is a sufficiently regular minimizer, then certainly $A(u + tv) \geq A(u)$ for all $t \in \mathbb{R}$ and all admissible variations v which satisfy $v|_{\partial\Omega} = 0$. The latter condition is required to ensure that $u + tv$ again is tied to the frame at the boundary. Let us denote by $a(w) = \sqrt{1 + |w|^2}$, i.e., w stands for ∇u in the formula for the surface area. Then by Taylor expansion, we see that $a(w + \delta w) = a(w) + a'(w) \cdot \delta w + O(|\delta w|^2)$ with derivative (=Jacobian) $a'(w) = \frac{w}{\sqrt{1 + |w|^2}}$. With this, we can formally compute

$$\begin{aligned} \delta A(u)[v] &:= \lim_{t \rightarrow 0} \frac{1}{t} (A(u + tv) - A(u)) \\ &= \lim_{t \rightarrow 0} \frac{1}{t} \int_{\Omega} (a(\nabla u + t\nabla v) - a(\nabla u)) dx = \int_{\Omega} a'(\nabla u) \cdot \nabla v dx \\ &= \int_{\Omega} \frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \cdot \nabla v dx \quad \forall v : v|_{\partial\Omega} = 0. \end{aligned}$$

This is called the *first variation* of the functional $A(u)$ in direction v , and corresponds to a directional derivative of the map $A : u \mapsto A(u)$. By our previous considerations, we see that $\delta A(u)[v] = 0$ for all admissible v , and hence

$$\int_{\Omega} \frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \cdot \nabla v = 0 \quad \forall v : v|_{\partial\Omega} = 0.$$

Since the test function v denotes a variation of u , this identity is called a *variational equation*, and likewise the corresponding minimization problem is called a

variational problem. Such infinite-dimensional minimization problems are studied in the context of the *calculus of variations*.

Let us proceed by recalling the integration-by-parts formula

$$\int_{\Omega} \vec{q} \cdot \nabla v \, dx = - \int_{\Omega} \operatorname{div}(\vec{q}) v \, dx + \int_{\partial\Omega} \vec{n} \cdot \vec{q} v \, ds,$$

which follows from the Gauß-integral law and the chain rule; see exercises. This allows us to rewrite the variational equation into the equivalent form

$$- \int_{\Omega} \operatorname{div} \left(\frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \right) v \, dx = 0 \quad \forall v : v|_{\partial\Omega} = 0.$$

Note that the boundary term in the integration-by-parts formula vanishes, since we have assumed $v|_{\partial\Omega} = 0$. To proceed further, we use the following observation.

Lemma 1.2 (Fundamental lemma 2).

Let $\Omega \subset \mathbb{R}^d$ be open and $f \in C(\overline{\Omega})$. Then

$$\int_{\Omega} f v \, dx = 0 \quad \forall v \in C(\overline{\Omega}) : v|_{\partial\Omega} = 0 \quad \Leftrightarrow \quad f(x) = 0 \quad \forall x \in \Omega.$$

The proof is again left as an exercise. We note that the statement can again be generalized concerning the regularity of the functions f and v .

By applying this result to the above equation now immediately leads to the nonlinear elliptic partial differential equation

$$- \operatorname{div} \left(\frac{1}{\sqrt{1 + |\nabla u|^2}} \nabla u \right) = 0 \quad \text{in } \Omega. \quad (1.6)$$

From the fact that the membrane is attached to the frame, we obtain the additional boundary condition

$$u = g \quad \text{on } \partial\Omega. \quad (1.7)$$

The system (1.6)–(1.7) describes the vertical position u of a membrane which is stretched over a rigid frame and which has the minimal surface area.

Linearization and the membrane problem

If $g \approx 0$, we have $u \approx 0$ and we may further assume the small-slope regime $|\nabla u| \ll 1$. As a consequence, we see that $\sqrt{1 + |\nabla u|^2} \approx 1$, and we obtain the corresponding linearized problem

$$\begin{aligned} -\Delta u &= 0 & \text{in } \Omega \\ u &= g & \text{on } \partial\Omega, \end{aligned}$$

which is called the *membrane equation* plus boundary conditions. Reverting the arguments used in the derivation of (1.6)–(1.7), one can show that solutions of this problem are minimizers of the quadratic energy

$$E(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx \rightarrow \min_{u: u|_{\partial\Omega} = g}.$$

Details are again left as an exercise. If we choose a right hand side f in the membrane equation and set $g = 0$ at the boundary, we arrive at our model problem (1.4)–(1.5) with $a = 1$. This system then models the deflection of a horizontal membrane under the action of a vertical distributed load f .

1.3 The biharmonic problem

The mechanical forces within a membrane are due to stretching and they act tangentially to the surface of the membrane. If we instead consider a thin plate, then the main forces are due to bending of the plate. Neglecting other effects and setting the material parameters to one, the deflection of a horizontal plate under the action of a vertical force f is described by the biharmonic equation

$$\Delta^2 u = f \quad \text{in } \Omega, \quad (1.8)$$

which is a linear partial differential equation of fourth order. This equation describes the equilibrium of internal and external forces in the *Kirchhoff-Love plate theory*; see Lecture: Continuum Mechanics. To fully describe the position of the plate, we additionally require appropriate boundary conditions, e.g.

$$u = 0 \quad \text{and} \quad \partial_n u = 0 \quad \text{on } \partial\Omega, \quad (1.9)$$

which describe that the plate is clamped. Note that we now have two conditions at every point of the boundary. Since by the first condition, also the tangential derivative along the boundary vanishes, we conclude that even $\nabla u = 0$ on $\partial\Omega$.

With very similar arguments as used in the derivation of the minimal surface and membrane equation, one can show that the equilibrium position u described by the system (1.8)–(1.9) corresponds to a minimum of

$$\frac{1}{2} \int_{\Omega} |\Delta u|^2 dx - \int_{\Omega} f u dx \rightarrow \min_{u: u|_{\partial\Omega} = \partial_n u|_{\partial\Omega} = 0}.$$

The equilibrium of forces is thus again tightly related to the minimization of a certain energy. Apart from the appearance of Δ instead of ∇ and the two boundary conditions instead of one, this problem has a very similar form as the membrane equation and as our model problem (1.4)–(1.5) and, in fact, can be treated with very similar arguments, both, analytically and numerically.

1.4 Linear elasticity

We now turn to the deformation of more general elastic bodies. In that case, the equilibrium of forces is expressed as

$$-\operatorname{div}(\sigma) = \vec{f} \quad \text{in } \Omega.$$

In the case of small deformations, we may identify $\Omega \subset \mathbb{R}^3$ with the reference configuration of the object under consideration. Furthermore, $\vec{f} : \Omega \rightarrow \mathbb{R}^3$ is the density of external forces, and $\sigma : \Omega \rightarrow \mathbb{R}^{3 \times 3}$ is a symmetric tensor field, the so-called stress tensor. The divergence operator acts in a row by row fashion, i.e., $\operatorname{div}(\sigma)_i = \operatorname{div}(\sigma_{i,\cdot})$. If we assume that the body behaves linearly elastically and that the deformation is small, the stresses can be related to the displacement field of the body $\vec{u} : \Omega \rightarrow \mathbb{R}^3$ by the Lamé equations

$$\sigma = 2\mu\varepsilon(\vec{u}) + \lambda \operatorname{div}(\vec{u})I.$$

These are a generalization of Hooke's law for an elastic spring. The coefficients μ and λ are the Lamé parameters, and the linearized strain tensor

$$\varepsilon(\vec{u}) = \frac{1}{2} (D\vec{u} + (D\vec{u})^\top).$$

describes the local stretching of the material. Here $(D\vec{u})_{ij} = \partial_j u_i$ is the Jacobian matrix of the displacement field $\vec{u}$, and hence $\varepsilon(\vec{u})$ amounts to a symmetrized gradient operator. Putting everything together, we arrive at the linear (better: linearized) elasticity equations

$$-\operatorname{div} (2\mu\varepsilon(\vec{u}) + \lambda \operatorname{div}(\vec{u})I) = \vec{f} \quad \text{in } \Omega. \quad (1.10)$$

This is a system of linear partial differential equations of second order. At the boundary, we may prescribe the displacement $\vec{u}$ or the normal forces, e.g., by

$$\vec{u} = \vec{g} \quad \text{on } \Gamma_D \quad (1.11)$$

$$\vec{n} \cdot \sigma = 0 \quad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \quad (1.12)$$

Using the relation between stresses and strains, the normal stress can be expressed in terms of the displacement $\vec{u}$. The boundary value problem (1.10)–(1.12) approximately describes the deformation $\vec{u}$ of a linearly elastic body under the action of an external load. Note that the equations are an approximation that is valid only for small deformations $\vec{u}$; for large deformations, the strain tensor $\varepsilon(\vec{u})$ has to be replaced by a nonlinear expression.

With similar arguments as used in the previous examples, one can see that the system (1.10)–(1.12) amounts to the Euler-Lagrange equations (=first order optimality conditions) for the variational problem

$$\frac{1}{2} \int_{\Omega} 2\mu|\varepsilon(\vec{u})|^2 + \lambda|\operatorname{div}(\vec{u})|^2 dx - \int_{\Omega} \vec{f} \cdot \vec{u} dx \rightarrow \min_{\vec{u} : \vec{u}|_{\Gamma_D} = \vec{g}}.$$

This can be verified with the very same arguments as used for the minimal surface and membrane problem. In the elasticity context, the test functions $\vec{v}$ appearing in this derivation are called *virtual displacements*. As we will see, the analytical and numerical tools developed for our model problem (1.4)–(1.5) will apply almost immediately also to the linear elasticity system. Further details will again be discussed in the Lecture: *Continuum Mechanics*.

Incompressible limit

For some particular materials, such as rubber, the parameter λ becomes extremely large. It is then convenient to introduce an auxiliary variable $p = -\lambda \operatorname{div}(\vec{u})$, which allows us to rewrite (1.10) as an equivalent system

$$\begin{aligned} -\operatorname{div}(2\mu\varepsilon(\vec{u})) + \nabla p &= \vec{f} & \text{in } \Omega \\ \operatorname{div}(\vec{u}) + \frac{1}{\lambda}p &= 0 & \text{in } \Omega. \end{aligned}$$

These equations now have a different structure than the elliptic problems considered before, and we will only briefly comment on the generalization of our arguments to such problems; details are discussed in the Lecture: *Numerical methods for Continuum Mechanics*. Note that the case of large λ now corresponds to a small perturbation of the limit problem, for which the second equation reads

$$\operatorname{div}(\vec{u}) = 0 \quad \text{in } \Omega.$$

This constraint restricts the set of possible deformations of an incompressible material. Note that $\int_{\omega} \operatorname{div}(\vec{u}) \, dx = 0$ for all $\omega \subset \Omega$, i.e., the first-order (infinitesimal) change in volume of every material subdomain vanishes. Exact volume preservation in the nonlinear theory would instead require $\det(I + D\vec{u}) = 1$.

1.5 Incompressible flow

The incompressible flow of a homogeneous Newtonian fluid is described by the famous Navier-Stokes equations

$$\begin{aligned} \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} - \nu \Delta \vec{u} + \nabla p &= \vec{f} \\ \operatorname{div}(\vec{u}) &= 0. \end{aligned}$$

Here $\vec{u}$ denotes the flow velocity, p is the pressure, and ν the kinematic viscosity. Under steady flow conditions and assuming slow flow velocities, the first two terms in the first equation can be dropped and we arrive at the Stokes system

$$-\nu \Delta \vec{u} + \nabla p = \vec{f} \quad \text{in } \Omega, \tag{1.13}$$

$$\operatorname{div}(\vec{u}) = 0 \quad \text{in } \Omega. \tag{1.14}$$

These equations describe the slow, steady and incompressible flow of a homogeneous Newtonian fluid. To complete the problem description, additional boundary conditions are required, e.g.,

$$\vec{u} = 0 \quad \text{on } \partial\Omega. \quad (1.15)$$

This condition means that the fluid perfectly sticks to the boundary and hence the velocity is zero there (=no slip condition).

Apart from the physical meaning of the variables $\vec{u}$ and p , the equations have exactly the same form as the incompressible limit of the linear elasticity system, and thus can be handled with the same analytical and numerical tools. With similar arguments as employed for the minimal surface problem, one can see, for instance, that solutions of (1.13)–(1.15) characterize the minima of

$$\frac{1}{2} \int_{\Omega} \nu |D\vec{u}|^2 dx - \int_{\Omega} \vec{f} \cdot \vec{u} dx \rightarrow \min_{\substack{\vec{u} \cdot \vec{n}|_{\partial\Omega}=0 \\ \operatorname{div}(\vec{u})=0}}.$$

The steady flow of an incompressible fluid and the deformation of an incompressible elastic material are thus again characterized by a variational problem encoding the principle of minimum energy. Details are discussed in the exercises.

1.6 Magneto-quasistatics

The propagation of electromagnetic waves is governed by Maxwell's equations. The two most important relations, due to Ampère and Faraday, are

$$\operatorname{curl} \vec{H} = \partial_t \vec{D} + \vec{J}, \quad \operatorname{curl} \vec{E} = -\partial_t \vec{B}.$$

These are balance laws for the electric and magnetic fields $\vec{E}$, $\vec{H}$, the corresponding fluxes $\vec{D}$, $\vec{B}$, also known as electric displacement and magnetic induction, and the current $\vec{J}$. Note that all these functions are vector valued!

To obtain a closed system, one again requires additional material laws. For a linear isotropic non-dispersive medium, these are given by

$$\vec{D} = \varepsilon \vec{E}, \quad \vec{B} = \mu \vec{H}, \quad \vec{J} = \vec{J}_s + \sigma \vec{E},$$

with permittivity ε , permeability μ and conductivity σ . The term $\vec{J}_s$ denotes the impressed currents, which act as sources for the electromagnetic fields.

By taking the divergence in Faraday's law and using $\operatorname{div} \operatorname{curl} = 0$, one can deduce that $\operatorname{div}(\partial_t \vec{B}) = 0$. Together with appropriate initial conditions, this

implies $\operatorname{div}(\vec{B}) = 0$. On a simply connected domain, the magnetic induction can thus be expressed in the form

$$\vec{B} = \operatorname{curl} \vec{A},$$

with $\vec{A}$ denoting a magnetic vector potential. Since $\operatorname{curl} \nabla \phi = 0$, we then also have $\vec{B} = \operatorname{curl}(\vec{A} - \nabla \phi)$ for all smooth scalar functions ϕ . We here assume that ϕ is chosen, such that $\vec{E} = -\partial_t \vec{A}$, which can again be justified via Faraday's law. With the help of these transformations and the material laws stated above, the remaining Amperère law can be rewritten as

$$\varepsilon \partial_{tt} \vec{A} + \sigma \partial_t \vec{A} + \operatorname{curl}(\mu^{-1} \operatorname{curl} \vec{A}) = \vec{J}_s.$$

In the magneto-quasistatic approximation, the first term in this equation, which corresponds to $\partial_t \vec{D}$, is neglected. After Laplace-transformation in time, we obtain

$$\kappa \vec{a} + \operatorname{curl}(\mu^{-1} \operatorname{curl} \vec{a}) = \vec{j}_s \quad \text{in } \Omega, \quad (1.16)$$

with $\kappa = s\sigma$, $\vec{a} = \mathcal{L}(\vec{A})$, and $\vec{j}_s = \mathcal{L}(\vec{J}_s)$. This is a second order elliptic partial differential equation, parametrized by the Laplace-parameter s . The corresponding homogeneous Dirichlet and Neumann boundary condition now read

$$\vec{n} \times \vec{a} = 0 \quad \text{on } \Gamma_D \quad (1.17)$$

$$\vec{n} \times (\mu^{-1} \operatorname{curl} \vec{a}) = 0 \quad \text{on } \partial\Omega \setminus \Gamma_D. \quad (1.18)$$

An appropriate weak formulation for this problem will be derived later.

The two-dimensional case

A setting of practical relevance, e.g. for electric machine simulation, is that $\Omega = \omega \times (0, L)$ and the magnetic and electric fields have the form $\vec{B} = (B_x, B_y, 0)$ and $\vec{E} = (0, 0, E_z)$, with component functions independent of z . We deduce that $\vec{a} = (0, 0, u)$ and $\vec{j}_s = (0, 0, f)$ with u and f depending on x and y only. In this case, the system (1.16)–(1.18) can be transformed into

$$\begin{aligned} \kappa u - \operatorname{div}(\mu^{-1} \nabla u) &= f & \text{in } \omega, \\ u &= 0 & \text{on } \partial\omega. \end{aligned}$$

For $\kappa = 0$, this again corresponds to our model problem (1.4)–(1.5). All results obtained for this model problem therefore apply also to the magneto-quasistatic problem in the 2-dimensional setting. As we will see later, most of the results can also be generalized to the three dimensional problem (1.16)–(1.18). Details will be discussed in the Lecture: *Computational Electromagnetics*.

1.7 Concluding remarks

All problems mentioned above are derived from similar physical principles, i.e. conservation or balance laws, constitutive equations, and similar mathematical arguments. They often lead to systems of partial differential equations with similar structure. In many cases, the differential equations, more precisely: the corresponding boundary value problems, are related to some minimization principle. These observations will allow us to develop a rather general approach to treat all these problems, analytically and numerically, within a common framework.

Variational problems

We will mainly develop our ideas for the model problem

$$\begin{aligned} -\operatorname{div}(a\nabla u) &= f && \text{in } \Omega, \\ u &= 0 && \text{on } \partial\Omega, \end{aligned}$$

but later show, that the concepts generalize nicely to the other problems mentioned above. We may multiply the differential equation by a test function v , integrate over the domain and use the integration-by-parts formula, to see that

$$\begin{aligned} \int_{\Omega} f v \, dx &= \int_{\Omega} -\operatorname{div}(a\nabla u) v \, dx \\ &= \int_{\Omega} a\nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} a\partial_n u v \, ds. \end{aligned}$$

Here we used $\vec{n} \cdot (a\nabla u) = a(\vec{n} \cdot \nabla u) =: a\partial_n u$ in the boundary term. Any smooth solution of our model problem thus also satisfies the variational principle

$$\int_{\Omega} a\nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v : v|_{\partial\Omega} = 0. \quad (\text{VP})$$

Since only first derivatives appear in this equation, we will call this the *weak* or *variational form* of our boundary value problem. Let us note that additionally the condition $u|_{\partial\Omega} = 0$ still has to be required. The weak formulation of all other model problems will have a very similar structure, which suggests to consider such problems on a more abstract level.

2 Elliptic variational problems

In this chapter, we study linear variational problems of the abstract form

$$\text{Find } u \in V: \quad a(u, v) = \ell(v) \quad \forall v \in V, \quad (\text{VP})$$

where V is some Hilbert space, $a : V \times V \rightarrow \mathbb{R}$ is a bilinear form, and $\ell : V \rightarrow \mathbb{R}$ is a linear mapping. We always assume that V is a vector space over $\mathbb{R}$; the complex case can be considered with minor modifications. The scalar product and norm of V are denoted by $(\cdot, \cdot)_V$ and $\|\cdot\|_V$, and we write V^* for the dual space of V , which consists of all linear functionals $\ell : V \rightarrow \mathbb{R}$ with bounded norm

$$\|\ell\|_{V^*} := \sup_{0 \neq v \in V} \frac{|\ell(v)|}{\|v\|_V}.$$

The action of a linear functional can also be expressed as $\ell(v) = \langle \ell, v \rangle_{V^* \times V}$ via the duality product. As a consequence of these definitions, we have

$$\ell(v) = \langle \ell, v \rangle_{V^* \times V} \leq \|\ell\|_{V^*} \|v\|_V$$

which holds for all $\ell \in V^*$ and all $v \in V$, and resembles the Cauchy-Schwarz inequality $(u, v)_V \leq \|u\|_V \|v\|_V$. The following notions will be of importance.

Definition 2.1. The bilinear form $a : V \times V \rightarrow \mathbb{R}$ is called

- elliptic, if there exists $\alpha > 0$ such that $a(u, u) \geq \alpha \|u\|_V^2$ for all $u \in V$;
- continuous, if there is a $C_a > 0$ with $|a(u, v)| \leq C_a \|u\|_V \|v\|_V$ for all $u, v \in V$;
- symmetric, if $a(u, v) = a(v, u)$ for all $u, v \in V$.

Convention. We will always use the same symbols α and C_a for the ellipticity and continuity constants, when we require that $a(\cdot, \cdot)$ has one of these properties.

2.1 Symmetric variational problems

Most of the following material should be familiar from the Lectures: *Partial Differential Equations* and *Numerical methods for Differential Equations*. For completeness of the presentation, we recall the main theoretical results here.

Lemma 2.2 (Equivalent minimization problem).

Let $a : V \times V \rightarrow \mathbb{R}$ be symmetric, continuous, and non-negative, i.e., $a(u, u) \geq 0$ for all $u \in V$. Then (VP) is equivalent to the minimization problem

$$J(u) = \min_{w \in V} J(w) \quad \text{with } J(w) := \frac{1}{2}a(w, w) - \ell(w). \quad (\text{MP})$$

Proof. Using (bi)linearity of $a(\cdot, \cdot)$ and $\ell(\cdot)$, we see that

$$J(u + tv) = J(u) + t[a(u, v) - \ell(v)] + \frac{t^2}{2}a(v, v) \quad \forall t \in \mathbb{R} \ \forall v \in V.$$

($\Rightarrow$) If (VP) holds, i.e., $a(u, v) = \ell(v)$, then $J(u + tv) = J(u) + \frac{t^2}{2}a(v, v) \geq J(u)$, since $a(\cdot, \cdot)$ was assumed non-negative. Hence u is a minimizer.

($\Leftarrow$) We proceed by contradiction: Assume that $a(u, v) - \ell(v) =: c < 0$ (the same idea works for > 0). If $a(v, v) = 0$, we see $J(u + tv) = J(u) + tc < J(u)$ for all $t > 0$. If $a(v, v) > 0$, we set $t = -c/a(v, v)$ and see that

$$J(u + tv) = J(u) - \frac{c}{a(v, v)}c + \frac{c^2}{2a(v, v)^2}a(v, v) = J(u) - \frac{c^2}{2a(v, v)} < J(u),$$

which is a contradiction to u being a minimizer. $\square$

As a next step, we show existence of a minimizer for (MP), which by the previous consideration amounts to a solution of the variational problem (VP).

Lemma 2.3 (Lax-Milgram, symmetric version).

Let $a : V \times V \rightarrow \mathbb{R}$ be elliptic, continuous, and symmetric. Then for any $\ell \in V^*$, the variational problem (VP) has a unique solution $u \in V$ and $\|u\|_V \leq \frac{1}{\alpha}\|\ell\|_{V^*}$.

Proof. We proceed as in the proof of the Riesz representation theorem, which is the special case $a(u, v) := (u, v)_V$; see the Lecture: *Functional Analysis*.

(i) From the properties of $a(\cdot, \cdot)$ and $\ell(\cdot)$, we see that

$$\begin{aligned} J(u) &= \frac{1}{2}a(u, u) - \ell(u) \geq \frac{\alpha}{2}\|u\|_V^2 - \|\ell\|_{V^*}\|u\|_V \\ &\geq \frac{\alpha}{2}(\|u\|_V - \frac{1}{\alpha}\|\ell\|_{V^*})^2 - \frac{1}{2\alpha}\|\ell\|_{V^*}^2 \geq -\frac{1}{2\alpha}\|\ell\|_{V^*}^2. \end{aligned}$$

This shows that $J(\cdot)$ is bounded from below and hence there exists $c := \inf_{v \in V} J(v)$.

(ii) By definition of the infimum, we can find a (minimizing) sequence $(u_n)_{n \in \mathbb{N}}$ such that $c \leq J(u_n) \leq c + \frac{1}{n}$. Using ellipticity and bilinearity of a , we then get

$$\begin{aligned} \alpha\|u_n - u_m\|_V^2 &\leq a(u_n - u_m, u_n - u_m) \\ &\stackrel{(*)}{=} 2a(u_n, u_n) + 2a(u_m, u_m) - a(u_n + u_m, u_n + u_m) \\ &= 4J(u_n) + 4J(u_m) - 8J\left(\frac{u_n + u_m}{2}\right) \\ &\leq 4\left(c + \frac{1}{n}\right) + 4\left(c + \frac{1}{m}\right) - 8c = 4\left(\frac{1}{n} + \frac{1}{m}\right). \end{aligned}$$

This shows that $(u_n)_{n \in \mathbb{N}}$ is a Cauchy sequence. The identity (*) used in the second step is called *parallelogram identity*.

(iii) Since V is a Hilbert space, and hence complete, there exists a limit $u := \lim_n u_n$, and by continuity of the functional $J(u)$, we see that

$$c = \lim_{n \rightarrow \infty} J(u_n) = J(\lim_{n \rightarrow \infty} u_n) = J(u),$$

which shows that u is a minimizer. By equivalence of the minimization (MP) with the variational problem, we see that u also is a solution of (VP).

(iv) Using ellipticity of $a(\cdot, \cdot)$ and continuity of $\ell(\cdot)$, we further see that

$$\alpha \|u\|_V^2 \leq a(u, u) = \ell(u) \leq \|\ell\|_{V^*} \|u\|_V,$$

which yields the a-priori bound. By linearity of the problem (VP), the difference $w = u_1 - u_2$ of two potential solutions satisfies (VP) with $\ell = 0$, and hence the solution is unique; see exercises. $\square$

Let us discuss an immediate consequence of this result.

Corollary 2.4 (Riesz representation theorem).

For any $\ell \in V^$, there exists a unique $u_\ell \in V$ such that $\ell = (u_\ell, \cdot)_V$, i.e.,*

$$(u_\ell, v)_V = \ell(v) = \langle \ell, v \rangle_{V^* \times V} \quad \forall v \in V.$$

Moreover, the mapping $R_V : V^ \rightarrow V$, $\ell \mapsto u_\ell$, is linear, continuous, and continuously invertible with $\|R_V \ell\|_V = \|\ell\|_{V^*}$ for all $\ell \in V^*$ and is called the Riesz isomorphism.*

For later reference, let us mention another direct consequence of Lemma 2.3.

Corollary 2.5 (Projection lemma).

Let $W \subset V$ be a closed subspace of the Hilbert space V . Then for any $u \in V$ there exists a unique element $u_W \in W$ such that

$$\|u - u_W\|_V = \min_{w \in W} \|u - w\|_V. \quad (2.1)$$

Moreover, the minimizer is characterized by

$$(u - u_W, w)_V = 0 \quad \forall w \in W. \quad (2.2)$$

Proof. The space W equipped with the norm $\|\cdot\|_V$ is again a Hilbert space. By squaring all terms, we see that

$$\frac{1}{2}\|u - w\|_V^2 = \frac{1}{2}(w, w)_V - (u, w)_V + \frac{1}{2}\|u\|_V^2.$$

Since $u \in V$ is assumed known here, the last term is constant, and the above minimization problem with $a(u, v) := (u, v)_V$ is thus equivalent to

$$\frac{1}{2}a(w, w) - \ell(w) \rightarrow \min_{w \in W}$$

with $a(w, w) = (w, w)_V$ symmetric, elliptic and continuous, and $\ell(w) = (u, w)_V$ linear and continuous. Existence and uniqueness of the minimizer now follows from Lemma 2.3 and the corresponding variational problem can be rephrased as

$$0 = \ell(w) - a(u_W, w) = (u, w)_V - (u_W, w)_V = (u - u_W, w)_V \quad \forall w \in W,$$

which already proves the second claim. $\square$

Remark 2.6. The element $u_W \in W$ in the above lemma is the best-approximation for $u \in V$ in the closed subspace $W \subset V$. The variational equation (2.2) states that the approximation error $u - u_W$ is orthogonal to the approximating subspace W ; draw a sketch! This is a special instance of *Galerkin orthogonality*, discussed in more detail later on. By the Pythagoras theorem, we further get

$$\begin{aligned} \|u\|_V^2 &= \|u_W + (u - u_W)\|_V^2 \\ &= \|u_W\|_V^2 + 2(u_W, u - u_W)_V + \|u - u_W\|_V^2 = \|u_W\|_V^2 + \|u - u_W\|_V^2. \end{aligned}$$

Any function $u \in V$ can thus be split orthogonally into $u = u_W + (u - u_W)$ with $u_W \in W$ and $u - u_W \in W^\perp$, which is the orthogonal complement of W in V .

2.2 The lemma of Lax-Milgram

A main observation of Peter Lax and Arthur Milgram was that the symmetry of the bilinear form is not required to obtain a unique solution of (VP).

Lemma 2.7 (Lax-Milgram, 1954).

Let $a : V \times V \rightarrow \mathbb{R}$ be an elliptic and continuous bilinear form. Then for any $\ell \in V^$, the problem (VP) has a unique solution $u \in V$ and $\|u\|_V \leq \frac{1}{\alpha}\|\ell\|_{V^*}$.*

Proof. The mapping $A : V \rightarrow V^*$, $u \mapsto a(u, \cdot)$ defines a bounded linear operator with $\|Au\|_{V^*} \leq C_a \|u\|_V$. Moreover, the variational problem (VP) is equivalent to the operator equation (details are verified in the exercises)

$$Au = \ell \quad \text{in } V^*. \quad (\text{OP})$$

For any $\omega \neq 0$, this is equivalent to the fixed-point equation

$$u = u + \omega R_V(\ell - Au) =: \Phi_\omega(u) \quad \text{in } V, \quad (\text{FP})$$

where $R_V : V^* \rightarrow V$ is the Riesz isomorphism of the Hilbert space V . This operator is needed to lift the residual $r := \ell - Au \in V^*$ back to the Hilbert space V ! The mapping $\Phi_\omega : V \rightarrow V$ is a self-map on V , and we have

$$\begin{aligned} \|\Phi_\omega(u) - \Phi_\omega(v)\|_V^2 &= \|(u - v) - \omega R_V A(u - v)\|_V^2 \\ &= \|u - v\|_V^2 - 2\omega(u - v, R_V A(u - v))_V + \omega^2 \|R_V A(u - v)\|_V^2. \end{aligned}$$

From the properties of the operators R_V and A , we see that

$$\begin{aligned} (u - v, R_V A(u - v))_V &= \langle A(u - v), u - v \rangle_{V^* \times V} \\ &= a(u - v, u - v) \geq \alpha \|u - v\|_V^2, \end{aligned}$$

where we used the ellipticity of $a(\cdot, \cdot)$. From its continuity, we further get

$$\|R_V A(u - v)\|_V^2 \leq \|A(u - v)\|_{V^*}^2 \leq C_a^2 \|u - v\|_V^2.$$

In summary, we thus obtain

$$\|\Phi_\omega(u) - \Phi_\omega(v)\|_V^2 \leq (1 - 2\alpha\omega + C_a^2\omega^2) \|u - v\|_V^2 =: L(\omega)^2 \|u - v\|_V^2.$$

With simple calculations, one can verify that $L(\omega)^2 < 1$ for all $0 < \omega < \frac{2\alpha}{C_a^2}$; see exercises. For such ω , the mapping Φ_ω is thus a contraction, and by Banach's fixed-point theorem, we obtain the existence of a unique solution to (FP). By equivalence of the problems, this then also is the unique solution of the operator equation (OP) and of the variational problem (VP). $\square$

2.3 Galerkin approximation and Cea's lemma

We next discuss the systematic approximation of elliptic variational problems.

Problem 2.8 (Galerkin approximation, 1915).

Let $V_h \subset V$ be some closed (e.g. finite-dimensional) subspace.

$$\text{Find } u_h \in V_h: \quad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h. \quad (\text{VP}_h)$$

Under the assumptions of the Lax-Milgram lemma, we immediately obtain a unique *discrete* solution $u_h \in V_h$ and $\|u_h\|_V \leq \frac{1}{\alpha} \|\ell\|_{V^*}$.

Remark 2.9 (Galerkin-isomorphism).

Let $\dim(V_h) = N$ and $\{\phi_1, \dots, \phi_N\} \subset V_h$ denote a basis of V_h . Then any element $u_h \in V_h$ can be expanded in this basis and

$$G_h : \mathbb{R}^N \rightarrow V_h, \quad \underline{u} \mapsto \sum_i \underline{u}_i \phi_i$$

is a linear and bijective mapping, the so-called *Galerkin isomorphism*. Moreover, the variational problem (VP_h) is equivalent to the linear system

$$\underline{A} \underline{u} = \underline{b} \quad \text{in } \mathbb{R}^N, \quad (2.3)$$

with $\underline{A}_{ij} = a(\phi_j, \phi_i)$ and $\underline{b}_i = \ell(\phi_i)$; see exercises and compare with (OP) in the proof of the Lax-Milgram lemma. From the ellipticity of $a(\cdot, \cdot)$, one can further deduce that $\underline{A}$ is positive definite, and hence regular.

We thus know that the Galerkin approximation u_h to u is well-defined and, in principle, even how to compute it. A natural next question is, of course, how well it approximates the true solution. This is answered by the following result.

Lemma 2.10 (Jean Céa, 1967, PhD thesis).

Let $a : V \times V \rightarrow \mathbb{R}$ be elliptic and continuous. Then

$$\|u - u_h\|_V \leq \frac{C_a}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Proof. By ellipticity and continuity of the bilinear form, and the definition of the continuous and discrete solution, we obtain

$$\begin{aligned} \alpha \|u - u_h\|_V^2 &\leq a(u - u_h, u - u_h) \\ &\stackrel{(\text{GO})}{=} a(u - u_h, u - v_h) \leq C_a \|u - u_h\|_V \|u - v_h\|_V. \end{aligned}$$

In the second step, we used Galerkin orthogonality, i.e. the fact that

$$a(u - u_h, w_h) = 0 \quad \text{for all } w_h \in V_h, \quad (\text{GO})$$

which follows immediately by subtracting the variational characterizations of the continuous and discrete solutions. The estimate then follows by dividing by the common factor and taking the infimum on both sides. $\square$

The Symmetric case and the Ritz method

If $a(\cdot, \cdot)$ is also symmetric, then the discrete variational problem (VP_h) is equivalent to the discrete minimization problem

$$\frac{1}{2}a(u_h, u_h) - \ell(u_h) \rightarrow \min_{u_h \in V_h}.$$

Using the Galerkin isomorphism $u_h = \sum_i \underline{u}_i \phi_i$, this can also be phrased as

$$\min_{\underline{u} \in \mathbb{R}^N} \frac{1}{2} \underline{u}^\top \underline{A} \underline{u} - \underline{u}^\top \underline{b},$$

i.e., as a quadratic minimization problem in $\mathbb{R}^N$. Standard optimization algorithms, e.g. the conjugate gradient method, can be used for its solution.

Walter Ritz used such problems before 1910 to compute approximate solutions (by hand!) for the biharmonic equation and similar variational problems. In the symmetric case, the bilinear form $a(u, v)$ defines a scalar product on V , and $\|u\|_a = \sqrt{a(u, u)}$ is the corresponding norm. From the projection lemma and the Pythagorean theorem, we then immediately obtain

$$\|u - u_h\|_a \leq \|u - v_h\|_a \quad \forall v_h \in V_h.$$

In the symmetric case, the Galerkin approximation u_h is thus the best approximation to u in the space V_h with respect to the energy norm. Together with the ellipticity and continuity of the bilinear form, this allows us to show that

$$\|u - u_h\|_V \leq \sqrt{\frac{C_a}{\alpha}} \inf_{v_h \in V_h} \|u - v_h\|_V$$

which is somewhat tighter than the estimate from Cea's lemma.

2.4 Density argument

Before closing this section, let us state an auxiliary result, which will be used intensively in the following chapter. Recall that a subspace $S \subset V$ is called *dense* in V if any element $u \in V$ can be approximated well by elements in S , i.e.,

$$\forall u \in V \quad \forall n \in \mathbb{N} \quad \exists s_n \in S : \|u - s_n\|_V \leq 1/n.$$

A linear operator $A : V \rightarrow W$ between normed spaces is called bounded if

$$\|A\|_{V \rightarrow W} := \sup_{0 \neq v \in V} \frac{\|Av\|_W}{\|v\|_V} < \infty.$$

We then immediately obtain the following result.

Lemma 2.11 (Closure principle, density argument).

Let $A : S \subset V \rightarrow W$ be a densely defined bounded linear operator between Hilbert (or Banach) spaces V, W . Then there exists a unique bounded linear operator $\bar{A} : V \rightarrow W$, called the extension or closure of A , with $\bar{A}s = As$ for all $s \in S$. Moreover, $\|\bar{A}\|_{V \rightarrow W} \leq \|A\|_{V \rightarrow W}$.

Proof. Let $u \in V$ be arbitrary and $(s_n)_{n \in \mathbb{N}} \subset S$ such that $\|u - s_n\|_V \leq 1/n$. Since A is linear and bounded, we see that

$$\|As_n - As_m\|_W \leq \|A\|_{V \rightarrow W} \|s_n - s_m\|_V.$$

This shows that the images $w_n = As_n$ define a Cauchy sequence in the complete space W . We may thus simply set $\bar{A}u := \lim_n As_n$. The same argument shows that the value is independent of the approximating sequence; see exercises. $\square$

3 Sobolev spaces

In the following, we introduce the function spaces that are of relevance for the weak formulation of elliptic partial differential equations. Details can be found in Evans: *Partial differential equations* and the Lecture: *Functional Analysis*.

Preliminaries and notation

Throughout this section, we let $\Omega \subset \mathbb{R}^d$ denote some bounded *domain*, i.e., a bounded, *open and connected set*. We denote by $L^p(\Omega)$, $1 \leq p \leq \infty$, the Lebesgue spaces of measurable functions on Ω with bounded norm

$$\|u\|_{L^p(\Omega)} = \left(\int_{\Omega} |u|^p dx \right)^{1/p}, \quad 1 \leq p < \infty,$$

and $\|u\|_{L^\infty(\Omega)} = \text{ess sup}_{x \in \Omega} |u(x)|$. Let us recall that $L^p(\Omega)$ actually consists of equivalence classes of functions that only differ on sets of measure zero and hence cannot be distinguished by the norms given above.

For all $1 \leq p \leq \infty$, the spaces $L^p(\Omega)$ are complete and $L^2(\Omega)$ is a Hilbert space with scalar product $(u, v)_{L^2(\Omega)} = \int_{\Omega} uv \, dx$. For $p < \infty$, the spaces $L^p(\Omega)$ are separable, i.e., they contain a countable dense set. Also recall Hölder's inequality

$$\left| \int_{\Omega} uv \, dx \right| \leq \|u\|_{L^p} \|v\|_{L^q}, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

which holds for all functions $u \in L^p(\Omega)$ and $v \in L^q(\Omega)$ with the corresponding indices. For $p = q = 2$, we obtain the Cauchy-Schwarz inequality.

3.1 Sobolev spaces

We denote by $C_0^\infty(\Omega)$ the set of infinitely differentiable functions with compact support, i.e., functions ϕ that vanish outside a compact set $K_\phi \subset \Omega$. Since the domain Ω was assumed open, any function $\phi \in C_0^\infty(\Omega)$ thus vanishes identically, i.e., together with all its derivatives, in the vicinity of the boundary $\partial\Omega$.

Distributional and weak derivative

For smooth functions u and multi-index $\alpha \in \mathbb{N}_0^d$, we denote by $D^\alpha u = \partial_1^{\alpha_1} \cdots \partial_d^{\alpha_d} u$ the classical derivatives of order α . Using integration-by-parts, we see that

$$\int_{\Omega} D^\alpha u \phi \, dx = (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi \, dx \quad \forall \phi \in C_0^\infty(\Omega).$$

All boundary terms vanish here, since ϕ has compact support. The right hand side of this formula is well-defined for any locally integrable function u and can be used to define the notion of a derivative for this rather large class of functions.

Definition 3.1 (Distributional and weak derivative).

(i) Let $L_{\text{loc}}^1(\Omega)$ denote the set of locally integrable functions, i.e., all measurable functions u with $u|_K \in L^1(K)$ for all compact subsets $K \subset \Omega$. Then for any function $u \in L_{\text{loc}}^1(\Omega)$, the linear mapping $T_u^\alpha : C_0^\infty(\Omega) \rightarrow \mathbb{R}$ defined by

$$\langle T_u^\alpha, \phi \rangle := (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi \, dx, \quad \phi \in C_0^\infty(\Omega)$$

is called the *distributional derivative* (of index α) of the function u .

(ii) If there exists a function $g^\alpha \in L_{\text{loc}}^1(\Omega)$ such that T_u^α can be expressed as

$$\langle T_u^\alpha, \phi \rangle = \int_{\Omega} g^\alpha \phi \, dx \quad \forall \phi \in C_0^\infty(\Omega),$$

then T_u^α is called *regular*, and g^α is called the *weak derivative* of u . If all weak derivatives for $|\alpha| \leq k$ exist, then u is called *k-times weakly differentiable*.

Remark 3.2. (i) Distributional derivatives can be defined for an even larger class of so-called *generalized functions*, or distributions; see Lecture: PDEs. Any function $u \in L_{\text{loc}}^1(\Omega)$ can again be identified with a *regular* distribution. For any distribution, derivatives of arbitrary order always exist in *distributional sense*.

(ii) A function $u \in C^k(\Omega)$ is also *k-times weakly differentiable* and the weak derivatives coincide with the usual derivatives. We therefore use the symbol $D^\alpha u$ also for the weak derivatives. Piecewise smooth functions with certain global regularity are also weakly differentiable and the weak derivative coincides with the piecewise classical derivative. Piecewise smooth functions with a genuine jump across an interface, on the other hand, are not weakly differentiable. Changes of a representative on a set of measure zero do not affect weak differentiability. In a similar manner, we can define weak versions of ∇u , $\text{div } \vec{j}$, $\text{curl } \vec{j}$, and so on; details are discussed in the exercises.

Sobolev spaces

For any $k \geq 0$ and $1 \leq p \leq \infty$, we can now define the Sobolev spaces

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \quad \forall \alpha : |\alpha| \leq k\}.$$

These are complete, i.e., Banach spaces, when equipped with the norms

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, \quad 1 \leq p < \infty,$$

and $\|u\|_{W^{k,\infty}(\Omega)} = \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}$ in the special case $p = \infty$, respectively. For $p = 2$, the Sobolev spaces are again Hilbert spaces and to emphasize this, we write $H^k(\Omega) = W^{k,2}(\Omega)$. We will also make use of the semi-norms

$$|u|_{W^{k,p}(\Omega)} = \left(\sum_{|\alpha|=k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, \quad 1 \leq p < \infty,$$

For $p = \infty$, we set $|u|_{W^{k,\infty}(\Omega)} = \max_{|\alpha|=k} \|D^\alpha u\|_{L^\infty(\Omega)}$. These seminorms take into account only the derivatives of highest order. For further details, see Evans: *Partial Differential Equations* or the Lecture: *Functional Analysis*.

Piecewise smooth functions

We already know that any smooth function $u \in C^k(\overline{\Omega})$ lies in $W^{k,p}(\Omega)$. The following result is a slight but important generalization of this observation.

Lemma 3.3 ($W^{k,p}$ -continuity).

Let $\Omega_h = \{\Omega_1, \dots, \Omega_N\}$ denote a non-overlapping partition of the domain Ω into (Lipschitz) regular sub-domains Ω_n , i.e., $\overline{\Omega} = \bigcup_n \overline{\Omega}_n$ and $\Omega_i \cap \Omega_j = \emptyset$ for $i \neq j$. Further denote by

$$C_{\text{pw}}^k(\Omega_h) := \{v \in L^p(\Omega) : v|_{\Omega_n} \in C^k(\overline{\Omega}_n), \quad 1 \leq n \leq N\}$$

the space of piecewise smooth functions defined on this partition, *where membership of the restriction in $C^k(\overline{\Omega}_n)$ means that it admits such an extension to the closure*. Then, for $k \geq 1$ and $v \in C_{\text{pw}}^k(\Omega_h)$,

$$v \in W^{k,p}(\Omega) \quad \Leftrightarrow \quad v \in C^{k-1}(\Omega).$$

Proof. We only consider the case $\overline{\Omega} = \overline{\Omega}_1 \cup \overline{\Omega}_2$ of two subdomains and $k = 1$. Writing $v_i := v|_{\Omega_i}$, the only reasonable candidate for the derivative is

$$g = \begin{cases} \nabla v_1, & x \in \Omega_1 \\ \nabla v_2, & x \in \Omega_2 \end{cases}$$

By inserting this into the definition of the distributional derivative and summing up over the partial derivatives, we have

$$\begin{aligned}
\langle \nabla v, \vec{\psi} \rangle &:= - \int_{\Omega} v \operatorname{div}(\vec{\psi}) dx \\
&= \int_{\Omega_1} \nabla v_1 \cdot \vec{\psi} dx + \int_{\Omega_2} \nabla v_2 \cdot \vec{\psi} dx - \int_{\partial\Omega_1} v_1 \vec{\psi} \cdot \vec{n}_1 ds - \int_{\partial\Omega_2} v_2 \vec{\psi} \cdot \vec{n}_2 ds \\
&= \int_{\Omega} g \cdot \vec{\psi} dx - \int_{\Gamma} (v_1 - v_2) \vec{\psi} \cdot \vec{n}_1 ds,
\end{aligned}$$

which holds for all $\vec{\psi} \in C_0^\infty(\Omega)^d$. In the last step, we used that $\vec{\psi}$ vanishes on $\partial\Omega$ and introduced $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ for the interface. The interface term vanishes for all $\vec{\psi}$ if and only if $v_1 - v_2 = 0$ on Γ ; compare with the fundamental lemma. The general case $k > 1$ and $N > 2$ can be treated by induction. $\square$

Since polynomials are infinitely differentiable, the above result characterizes the required continuity for piecewise polynomials to lie in certain Sobolev spaces. This will become important for the construction of finite element spaces later on.

3.2 Density, embedding and traces

By construction, the Sobolev spaces again consist of equivalence classes of functions that differ only on a set of measure zero, and it seems difficult to work with these objects. The following properties allow us to deal with this issue and to calculate with elements of Sobolev spaces as with ordinary functions.

Lemma 3.4 (Density).

Let $1 \leq p < \infty$ and $m \geq k$. Then the following assertions hold:

- (i) If $\Omega \subset \mathbb{R}^d$ is open, then the spaces $C^m(\Omega) \cap W^{k,p}(\Omega)$ are dense in $W^{k,p}(\Omega)$.
- (ii) If Ω is a bounded Lipschitz domain, then even $C^m(\overline{\Omega}) \subset W^{k,p}(\Omega)$ is dense.

The proof works by constructing an approximating sequence $\{\phi_n\}_{n \in \mathbb{N}} \subset C^\infty(\Omega)$ via so-called *mollification*; see Evans: *Partial Differential Equations* for details.

Remark 3.5. A domain $\Omega \subset \mathbb{R}^d$ is called a Lipschitz domain if there exists

- (a) a finite collection $(U_n)_{n \leq N}$ of open sets, such that $\partial\Omega \subset \bigcup_{n \leq N} U_n$;
- (b) bi-Lipschitz functions $f_n : O_n \subset \mathbb{R}^d \rightarrow U_n$ (Lipschitz, bijective, with Lipschitz inverse) such that $\partial\Omega \cap U_n = f_n(O_n \cap \{x_d = 0\})$ and $\Omega \cap U_n = f_n(O_n \cap \{x_d < 0\})$.

Make a sketch to visualize the meaning of this definition!

Trace theorem

As a first application of the density argument, we show that one can make sense of boundary values of a Sobolev function.

Lemma 3.6 (Trace theorem).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $1 \leq p < \infty$. Then there exists a unique bounded linear operator $\gamma : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$, such that $\gamma u = u|_{\partial\Omega}$ for all $u \in C^1(\overline{\Omega})$ and

$$\|\gamma u\|_{L^p(\partial\Omega)} \leq C_{\text{tr}} \|u\|_{W^{1,p}(\Omega)} \quad \forall u \in W^{1,p}(\Omega)$$

with C_{tr} depending only on p and the Lipschitz character and domain size.

Proof. To illustrate the basic arguments, we sketch the proof for the special case $p = 2$ and $\Omega = (a, b) \times (c, d)$; the general case can be obtained similarly and by decomposition and mapping of the boundary; see exercises. We denote the top part of the boundary by $\Gamma_{\text{top}} := (a, b) \times \{d\}$. Since $C^1(\overline{\Omega}) \subset H^1(\Omega)$ is dense, the trace operator γ is densely defined and linear. We next show the uniform bound for $u \in C^1(\overline{\Omega})$. By the fundamental theorem of calculus, we have

$$u(x, d) = u(x, y) + \int_y^d \partial_y u(x, \eta) d\eta.$$

Squaring and application of the Young and Cauchy-Schwarz inequalities yields

$$\begin{aligned} |u(x, d)|^2 &\leq 2|u(x, y)|^2 + 2|y - d| \int_y^d |\partial_y u(x, \eta)|^2 d\eta \\ &\leq 2|u(x, y)|^2 + 2|c - d| \int_c^d |\partial_y u(x, \eta)|^2 d\eta. \end{aligned}$$

By integrating over x and y on both sides, we then arrive at

$$\begin{aligned} |d - c| \int_a^b |u(x, d)|^2 dx &\leq 2 \int_c^d \int_a^b \left[|u(x, y)|^2 + |c - d| \int_c^d |\partial_y u(x, \eta)|^2 d\eta \right] dx dy \\ &= 2\|u\|_{L^2(\Omega)}^2 + 2|c - d|^2 \|\partial_y u\|_{L^2(\Omega)}^2. \end{aligned}$$

Hence we have shown that

$$\|u\|_{L^2(\Gamma_{\text{top}})}^2 \leq \frac{2}{|d - c|} \|u\|_{L^2(\Omega)}^2 + 2|c - d| \|\partial_y u\|_{L^2(\Omega)}^2 \leq C_{\text{tr}}^2 \|u\|_{H^1(\Omega)}^2.$$

In a similar manner, we can proceed with the other parts of the boundary, which leads to the desired trace estimate for smooth functions. Hence γ is a densely defined, linear, and bounded operator on $C^1(\overline{\Omega}) \subset H^1(\Omega)$ and can thus be extended to a bounded linear operator defined on the whole space $H^1(\Omega)$. $\square$

The proof of the trace estimate is constructive and the constant C_{tr} only depends on the Lipschitz character and size of the domain Ω . Let us further note that the trace estimate is not sharp, i.e., not every function $g \in L^p(\partial\Omega)$ is the trace of a function $u \in W^{1,p}(\Omega)$ for $1 < p < \infty$. For details, see the literature and the next chapter.

Embedding and extension

The trace lemma is, in fact, a special case of a so-called *embedding* result, which shows that differentiability always leads to some extra integrability; the reverse is not true! For later reference, we mention the most important results, whose proof can be found in the literature cited in the beginning of this chapter.

Lemma 3.7 (Embeddings, Sobolev, Rellich-Kondrachov).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $k \geq 1$. Then the embeddings

$$W^{k,p}(\Omega) \hookrightarrow \begin{cases} L^q(\Omega), & \text{for } p < \frac{d}{k} \text{ and } p \leq q \leq \frac{dp}{d-kp}, \\ L^q(\Omega), & \text{for } p = \frac{d}{k} \text{ and } p \leq q < \infty, \\ C(\overline{\Omega}), & \text{if } p > \frac{d}{k} \end{cases}$$

are continuous, i.e., $\|u\|_* \leq C_* \|u\|_{W^{k,p}(\Omega)}$ with appropriate constants C_* depending on the spaces and their canonical norms, but independent of u . Moreover

$$W^{k,p}(\Omega) \hookrightarrow \begin{cases} L^q(\Omega), & \text{for } p < \frac{d}{k} \text{ and } p \leq q < \frac{dp}{d-kp}, \\ L^q(\Omega), & \text{for } p = \frac{d}{k} \text{ and } p \leq q < \infty, \\ C(\overline{\Omega}), & \text{if } p > \frac{d}{k} \end{cases}$$

are compact and, in particular, $W^{k,p}(\Omega) \hookrightarrow W^{k-1,p}(\Omega)$ is compact.

By applying the results to $u = D^\alpha w$, we can formulate similar embedding results for $W^{k,p} \hookrightarrow W^{j,q}$ and $W^{k,p} \hookrightarrow C^j$ with $j < k$; see exercises and the given literature. For later reference, let us also mention the following useful result.

Lemma 3.8 (Extension).

Let $\Omega, \tilde{\Omega} \subset \mathbb{R}^d$ be bounded Lipschitz domains with $\overline{\Omega} \subset \tilde{\Omega}$. Then there exists a bounded linear operator $E : W^{k,p}(\Omega) \rightarrow W^{k,p}(\tilde{\Omega})$, $u \mapsto \tilde{u}$ such that $\tilde{u}|_\Omega = u$. Moreover, $\|\tilde{u}\|_{W^{k,p}(\tilde{\Omega})} \leq C \|u\|_{W^{k,p}(\Omega)}$ with C depending only on $\Omega, \tilde{\Omega}, k$ and p .

Without going into details, let us mention that the proofs of both results, continuous embedding and extension, are constructive, i.e., the constants C_* and C can actually be estimated from properties of the domain.

Calculus in Sobolev spaces

The density lemma implies that Sobolev functions can be approximated with arbitrary accuracy in the norm of the Sobolev space by smooth functions, i.e.,

$$W^{k,p}(\Omega) = \overline{C^m(\Omega) \cap W^{k,p}(\Omega)}^{\|\cdot\|_{W^{k,p}}} \quad \text{and} \quad W^{k,p}(\Omega) = \overline{C^m(\overline{\Omega})}^{\|\cdot\|_{W^{k,p}}},$$

if Ω is Lipschitz. This allows us to transfer many of the known formulas for smooth functions to Sobolev spaces, e.g., the integration-by-parts formula, the chain rule, etc., as long as all terms arising in the formulas are meaningful. Some examples in this direction will be discussed in the exercises.

3.3 Sobolev's norm-equivalence theorem

In the following, we present some results that allow us to define equivalent norms in Sobolev spaces. This will be an important ingredient for the analysis of variational problems later on. We start with an elementary observation, which motivates the following results.

Lemma 3.9. *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, $k \geq 1$, $u \in W^{k,p}(\Omega)$, and $D^\alpha u = 0$ for all $|\alpha| = k$. Then u is a polynomial of degree $\leq k - 1$.*

Proof. Let $B \Subset \Omega$ be an open ball and let $u_\varepsilon = \rho_\varepsilon * u$ be a mollification of u . By the standard properties of convolution and mollification in distribution theory, for sufficiently small $\varepsilon > 0$, we have $D^\alpha u_\varepsilon = \rho_\varepsilon * D^\alpha u = 0$ for every $|\alpha| = k$, and $u_\varepsilon \rightarrow u$ in $\mathcal{D}'(B)$. Since u_ε is smooth and B is convex, Taylor's theorem shows that $u_\varepsilon|_B$ is a polynomial of degree at most $k - 1$. Since the finite-dimensional space of polynomials of degree at most $k - 1$ is closed in $\mathcal{D}'(B)$, the restriction $u|_B$ is also a polynomial. Polynomials obtained on two overlapping balls coincide, since they agree on a nonempty open set. Finally, the connectedness of Ω allows us to connect any two balls by a finite chain of overlapping balls. Hence all local polynomials are restrictions of one polynomial of degree at most $k - 1$. $\square$

Up to a polynomial in $P_{k-1} = \text{span}\{x^\alpha : |\alpha| \leq k - 1\}$, a function $u \in W^{k,p}(\Omega)$ is thus determined already by its highest order derivatives $D^\alpha u$ for $|\alpha| = k$, but on P_{k-1} , all norms are equivalent. This motivates the following important result.

Lemma 3.10 (Sobolev norm-equivalence theorem).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $k \geq 1$, $1 \leq p < \infty$, and $\ell_j : W^{k,p}(\Omega) \rightarrow \mathbb{R}$, $1 \leq j \leq J$ be a sequence of bounded linear functionals such that

- (i) $\exists c_j: |\ell_j(u)| \leq c_j \|u\|_{W^{k,p}(\Omega)}$ for all $u \in W^{k,p}(\Omega)$;
(ii) $\ell_j(p_{k-1}) = 0$ for all $1 \leq j \leq J$ implies $p_{k-1} = 0$ for every $p_{k-1} \in P_{k-1}$.

Then $\|u\|_* := (\|u\|_{W^{k,p}(\Omega)}^p + \sum_{j=1}^J |\ell_j(u)|^p)^{1/p}$ defines a norm on $W^{k,p}(\Omega)$ and

$$c_* \|u\|_{W^{k,p}(\Omega)} \leq \|u\|_* \leq C_* \|u\|_{W^{k,p}(\Omega)} \quad \forall u \in W^{k,p}(\Omega)$$

with constant $c_*, C_* > 0$, i.e., the norms $\|\cdot\|_* \simeq \|\cdot\|_{W^{k,p}(\Omega)}$ are equivalent.

Proof. By using the previous lemma, it is easy to show that $\|\cdot\|_*$ defines a norm; see Exercises. Using (i), we further see that

$$\|u\|_*^p \leq \|u\|_{W^{k,p}}^p + \sum_{j=1}^J c_j^p \|u\|_{W^{k,p}}^p \leq (1 + \sum_j c_j^p) \|u\|_{W^{k,p}(\Omega)}^p,$$

which shows the upper bound with $C_* = (1 + \sum_j c_j^p)^{1/p}$. The lower bound is established by contradiction. If it were false, then for every $n \in \mathbb{N}$ we could choose a function $u_n \in W^{k,p}(\Omega)$ such that

$$\|u_n\|_{W^{k,p}(\Omega)} = 1 \quad \text{and} \quad \|u_n\|_* \leq \frac{1}{n}. \quad (\text{A})$$

Then from (A), we see that $\|u_n\|_* \leq \frac{1}{n}$, and hence

$$|u_n|_{W^{k,p}(\Omega)} \rightarrow 0 \quad \text{and} \quad \ell_j(u_n) \rightarrow 0 \quad \forall j.$$

By the compact embedding of $W^{k,p} \hookrightarrow W^{k-1,p}$, we further know that $\{u_n\}_{n \in \mathbb{N}}$ has a sub-sequence, again denoted by $\{u_n\}_{n \in \mathbb{N}}$, that converges in the norm of $W^{k-1,p}(\Omega)$ with limit $\bar{u} = \lim_n u_n \in W^{k-1,p}(\Omega)$. The convergence in $W^{k-1,p}(\Omega)$ and the estimate

$$|u_n - u_m|_{W^{k,p}(\Omega)} \leq |u_n|_{W^{k,p}(\Omega)} + |u_m|_{W^{k,p}(\Omega)} \rightarrow 0$$

show that this subsequence is a Cauchy sequence in $W^{k,p}(\Omega)$. By completeness, it converges to some limit in $W^{k,p}(\Omega)$, which must equal $\bar{u}$ by the already established convergence in $W^{k-1,p}(\Omega)$. Moreover, $D^\alpha \bar{u} = 0$ for all $|\alpha| = k$, and the previous lemma therefore yields $\bar{u} \in P_{k-1}$. Thus $u_n \rightarrow \bar{u}$ in $W^{k,p}(\Omega)$ and

$$\|\bar{u}\|_{W^{k,p}(\Omega)} = \|\lim_n u_n\|_{W^{k,p}(\Omega)} = \lim_n \|u_n\|_{W^{k,p}(\Omega)} = 1.$$

By the continuity of ℓ_j , on the other hand, we see that

$$\ell_j(\bar{u}) = \ell_j(\lim_n u_n) = \lim_n \ell_j(u_n) = 0,$$

and hence $\bar{u} \equiv 0$ by assumption (ii), which yields a contradiction. Hence (A) cannot be true, i.e., there exists a constant $c_* > 0$ for the lower bound. $\square$

Remark 3.11. The above result is very general but the proof, whose basic ingredient was *compactness*, is not constructive! In particular, the dependence of the constant c_* on the domain Ω is therefore not clear.

Example 3.12 (Poincaré-type inequalities).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Then

- (i) $\|u\|_{W^{1,p}(\Omega)} \simeq \|u\|_* := (|\int_V u(x)dx|^p + \|\nabla u\|_{L^p(\Omega)}^p)^{1/p}, \quad V \subset \Omega : |V| > 0;$
- (ii) $\|u\|_{W^{1,p}(\Omega)} \simeq \|u\|_* := (|\int_S u(x)ds|^p + \|\nabla u\|_{L^p(\Omega)}^p)^{1/p}, \quad S \subset \partial\Omega : |S| > 0;$
- (iii) $\|u\|_{W^{k,p}(\Omega)} \simeq \|u\|_* := (\|u\|_{L^p(\Omega)}^p + |u|_{W^{k,p}(\Omega)}^p)^{1/p}.$

The validity of these assertions will be verified in the exercises. For the last statement, a slight variation of the above proof is needed. Note that the value of the constants in the above norm equivalences is not obtained from these proofs.

3.4 The Friedrichs and Poincaré inequality

Let us now make some additional remarks on the function spaces, that will be relevant for our further considerations. By the closure principle, we can define

$$H_0^1(\Omega) = \overline{C_0^\infty(\Omega)}^{H^1(\Omega)},$$

which consists of all limits of smooth functions with compact support. By noting that $C_0^\infty(\Omega) \subset C^\infty(\Omega) \cap H^1(\Omega)$, we already know that $H_0^1(\Omega) \subset H^1(\Omega)$. Moreover, by construction the space $H_0^1(\Omega)$ is also complete.

Lemma 3.13 (Alternative characterization of $H_0^1(\Omega)$).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Then

- (i) $H_0^1(\Omega) = \{u \in H^1(\Omega) : u|_{\partial\Omega} = 0\}$, where $u|_{\partial\Omega} := \gamma u$ is the boundary trace of u .
- (ii) Any function $u \in H_0^1(\Omega)$ can be extended by zero to a function $\tilde{u} \in H_0^1(\mathbb{R}^d)$.

Remark 3.14. For any domain $\Omega \subset \mathbb{R}^d$, we may define $H_0^k(\Omega) = \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{H^k}}$. For Lipschitz domains, these spaces can again be characterized alternatively by

$$H_0^k(\Omega) = \{v \in H^k(\Omega) : D^\alpha v|_{\partial\Omega} = 0, |\alpha| \leq k-1\},$$

which for $k = 1$ follows like point (i) of the above lemma. Similar constructions are also possible for the spaces $W^{k,p}(\Omega)$ with vanishing boundary values.

Lemma 3.15 (Friedrichs inequality).

Let Ω be a bounded domain. Then, with $C_F = \text{diam}(\Omega)$, we have

$$\|u\|_{L^2(\Omega)} \leq C_F \|\nabla u\|_{L^2(\Omega)} \quad \forall u \in H_0^1(\Omega).$$

We further have $\|\nabla u\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)} \leq C'_F \|\nabla u\|_{L^2(\Omega)}$ for all $u \in H_0^1(\Omega)$, where $C'_F = \sqrt{1 + C_F^2}$.

The estimates with "some" constant C_F follow immediately from Sobolev's equivalence norm theorem; see Example 3.12(ii). The following proof, therefore, is intended to show that the constant can be chosen explicitly.

Proof. Set $a := \inf_{x \in \Omega} x_1$, $b := \sup_{x \in \Omega} x_1$, $L := b - a \leq \text{diam}(\Omega)$. For $u \in C_0^\infty(\Omega)$, let $\tilde{u} \in C_0^\infty(\mathbb{R}^d)$ be its extension by zero. Writing $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{d-1}$, the fundamental theorem of calculus and the Cauchy–Schwarz inequality give

$$|\tilde{u}(x_1, x')|^2 = \left| \int_a^{x_1} \partial_1 \tilde{u}(\xi, x') d\xi \right|^2 \leq L \int_a^{x_1} |\partial_1 \tilde{u}(\xi, x')|^2 d\xi.$$

Integration over $(a, b) \times \mathbb{R}^{d-1}$ therefore yields

$$\|u\|_{L^2(\Omega)}^2 = \|\tilde{u}\|_{L^2(\mathbb{R}^d)}^2 \leq L^2 \|\partial_1 \tilde{u}\|_{L^2(\mathbb{R}^d)}^2 \leq \text{diam}(\Omega)^2 \|\nabla u\|_{L^2(\Omega)}^2.$$

For arbitrary $u \in H_0^1(\Omega)$, choose $u_n \in C_0^\infty(\Omega)$ with $u_n \rightarrow u$ in $H^1(\Omega)$ and pass to the limit in this estimate. Finally,

$$\|u\|_{H^1(\Omega)}^2 = \|u\|_{L^2(\Omega)}^2 + \|\nabla u\|_{L^2(\Omega)}^2 \leq (1 + C_F^2) \|\nabla u\|_{L^2(\Omega)}^2,$$

which proves the second estimate with $C'_F = \sqrt{1 + C_F^2}$. $\square$

In a similar spirit, but with more involved arguments, one can show that also the constants in the Poincaré-type inequalities can be chosen depending on certain geometrical properties of the domain Ω . Let us state the following

Lemma 3.16 (Classical Poincaré inequality).

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $\bar{u} = \frac{1}{|\Omega|} \int_\Omega u dx$. Then

$$\|u - \bar{u}\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)} \quad \forall u \in H^1(\Omega).$$

If the domain Ω is star-shaped with respect to a ball $B \subset \Omega$, then C_P depends only on $\text{diam}(\Omega)$ and $1/\text{diam}(B)$. If Ω is convex, then $C_P \leq \text{diam}(\Omega)/\pi$.

Proof. The first assertion is a direct consequence of Sobolev's equivalent-norm theorem; see the exercises. The second can be found in Brenner & Scott: The Mathematical Theory of Finite Element Methods; called "Friedrichs' inequality" there. The third has been proven in a famous paper by Payne & Weinberger. $\square$

Remark 3.17 (Convex and star-shaped domains).

A domain $\Omega \subset \mathbb{R}^d$ is called convex if for any $x, \bar{x} \in \Omega$ the line $[x, \bar{x}]$ connecting the two points lies in Ω . It is called star-shaped with respect to the point $\bar{x} \in \Omega$ if any line $[x, \bar{x}]$ with $x \in \Omega$ lies in Ω . It is called star-shaped with respect to a ball $B \subset \Omega$ if it is star-shaped with respect to every $\bar{x} \in B$. Make a sketch!

Remark 3.18 (Poincaré vs. Taylor).

The Poincaré inequality states that any function $u \in H^1(\Omega)$ can be approximated by a constant function $\bar{u}$ such that the remainder depends only on the first derivative ∇u , which is a type of Taylor estimate. The extension to higher-order polynomial approximations is called the *Bramble-Hilbert lemma* and will be used and proved explicitly later to derive *interpolation error estimates*.